

Jaeseo Lee

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Summary

I specialize in ensuring software reliability and correctness through formal methods and system-level verification. My work focuses on building practical verification frameworks that enhance the safety and robustness of complex software systems. I have developed unified semantic models that integrate program logic with real-world system behaviors, enabling more realistic analysis of embedded and concurrent systems. To ensure scalability in real applications, I design and implement state-space reduction and optimization techniques that improve the efficiency of verification tools. I am particularly interested in applying formal reasoning and automated analysis to strengthen software quality in large-scale, safety-critical systems.

Education

Software Verification Lab. (POSTECH) <i>MS/Ph.D. in Computer Science and Engineering</i>	<i>Pohang, South Korea</i> <i>Feb 2017 - Present</i>
Pohang University of Science and Technology (POSTECH) <i>BS in Industrial and Management Engineering</i>	<i>Pohang, South Korea</i> <i>Mar 2011 - Feb 2017</i>
University of California, Berkeley <i>Concurrent Enrollment Program</i>	<i>Berkeley, California</i> <i>Jan 2015 – Dec 2015</i>
• Coursework: Operating Systems, Architecture, Machine Learning, Compiler, Security	

Industry Collaboration Projects

Verification on PLC Programs , with <i>KSOE (HD Korea Shipbuilding & Offshore Engineering Co., Ltd.)</i>	<i>Jan 2020 – Dec 2020</i>
• Clarified the ambiguous semantics of PLC language described in natural languages • Devised a bounded linear temporal logic (LTL) model checking method that checks conformity of PLC programs to specifications • Designed a specification language for expressing desired properties of PLC programs • Developed STBMC [tool] that integrates the whole process of PLC program verification. This tool generates a counterexample if and only if one exists	
Equivalence of LLVM IR Programs , with <i>GT One</i>	<i>June 2017 – Nov 2018</i>
• Machine-proved semantic equivalence of original and transformed code in security-enhancing transformations • Developed a lightweight tool with a translation validation approach	

Publications

Bounded Model Checking of Multitask PLC ST Programs with Preemption using Rewriting Modulo SMT (submitted) Jaeseo Lee, Kyungmin Bae	<i>JLAMP, 2025</i>
Formal Analysis of Networked PLC Controllers Interacting with Physical Environments Jaeseo Lee, Kyungmin Bae [paper]	<i>SAS, 2025</i>
Formal Semantics and Analysis of Multitask PLC ST Programs with Preemption Jaeseo Lee, Kyungmin Bae [paper]	<i>FM, 2024</i>
Bounded Model Checking of PLC ST Programs using Rewriting Modulo SMT Jaeseo Lee, Sangki Kim, Kyungmin Bae [paper]	<i>FTSCS, 2022</i>
Lightweight Equivalence Checking of Code Transformation for Code Pointer Integrity (in Korean) Jaeseo Lee, Tae-Hyoung Choi, Gyuho Lee, Jaegwan Yu, Kyungmin Bae [paper]	<i>KCSE, 2019.12</i>

Teaching

CSED332: Software Design Methods (TA)

Fall 2017, Fall 2019

CSED321: Programming Languages (TA)

Spring 2019

Scholarships

National Science & Technology Scholarship, by KOSAF (Kr. Student Aid Foundation)

Mar 2011 - Feb 2017

Additional Work Experience

NSW Department of Education

Sydney, Australia

Jan 2014 - Feb 2014

- Managed and digitized document workflows for efficient record-keeping
- Converted physical records to digital formats and organized signed forms for compliance
- Participated in departmental meetings to observe administrative and policy processes